

Eyad Arshad

AI STUDENT & SOFTWARE DEVELOPER

CONTACT

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TECHNICAL SKILLS

LANGUAGES

- C++
- Python
- JavaScript
- PHP
- HTML5
- CSS3

ENGINES & FRAMEWORKS

- Unreal Engine 5
- Qt 5/6
- Bootstrap
- SFML

AI & SECURITY

- scikit-learn
- YOLOv8
- OpenCV
- ONNX
- PE Analysis

DATABASES & TOOLS

- MySQL
- Firebase
- Git
- Arduino
- XAMPP

CONCEPTS

- OOP
- Multithreading
- Behavior Trees
- Procedural Gen
- RBAC
- IoT
- REST APIs

CERTIFICATIONS

HTML, CSS & JavaScript
IBM via Coursera · 2024

Web Development Course
NICON, Islamabad · 2019

INTERESTS

- Computer Vision
- Game AI
- IoT Systems
- Algorithms
- Full-Stack Dev

EDUCATION

Air University

Bachelor of Science — Artificial Intelligence

Relevant: Data Structures, OOP, Computer Networks, Database Systems, Machine Learning, Computer Vision

2024 – 2028
Islamabad, PK

PROJECTS

HELIX — AI-Powered Malware Detection System

github.com ↗

Python · PyQt6 · scikit-learn · pefile · x86 Emulation · Flask

- Built a hybrid malware detection engine combining 22 static PE features (import analysis, section entropy, Authenticode signatures) with 14 behavioral features from a custom x86 instruction emulator — 38-feature ML pipeline achieving 99.6% accuracy
- Designed distributed online learning architecture where user corrections sync to a central Flask server, retraining the model and propagating improved weights to all connected clients
- Implemented real-time background protection as a Windows system tray service monitoring Downloads, Desktop, and Temp directories with automatic threat alerts and quarantine
- Integrated VirusTotal API for hash-based threat intelligence and Authenticode certificate chain validation to reduce false positives on signed binaries

Smart Traffic Management System

github.com ↗

C++ · Qt · OpenCV · YOLOv8 · Arduino

- Engineered real-time traffic control using YOLOv8 (ONNX) for vehicle detection across multi-road intersections with 5-level density classification
- Designed multi-threaded Qt architecture separating camera processing from UI thread, enabling simultaneous feed analysis without frame drops
- Built adaptive signal timing algorithm dynamically adjusting green-light duration (5–25s) based on live vehicle density
- Integrated Arduino via QSerialPort for physical light control; automated violation detection saves timestamped evidence frames

Death Realm — First-Person Horror Game

github.com ↗

Unreal Engine 5 · C++ · Blueprints · Behavior Trees · NavMesh

- Developed complete first-person horror game with procedurally generated mazes using recursive backtracking, ensuring unique layouts each playthrough
- Built AI enemy system with UE5 Behavior Trees and NavMesh supporting patrol, chase, and attack states with dynamic difficulty scaling
- Engineered save/load system using UE's SaveGame API, persisting player state, level progress, and settings across sessions
- Architected modular C++ gameplay classes (MazeManager, MonsterAI, TrapCell) with Blueprint exposure for rapid design iteration

UtiliSOFT — Database Management System

github.com ↗

C++ · Qt 6 · MySQL · Qt Charts

- Built full-stack desktop ERP system for retail operations with 3-tier role-based access control (Owner / Manager / Salesman)
- Implemented secure authentication with session management and SQL injection prevention via Qt prepared statements
- Delivered 7 integrated modules — product catalog, stock control, sales processing, vendor, employee, debt tracking, and analytics dashboard

Enterprise Server & Cloud Storage System

Computer Networks · Client-Server Architecture

- Designed and implemented an industry-level client-server architecture with centralized cloud storage capabilities
- Engineered file transfer, authentication, and session management protocols as part of advanced Computer Networks coursework

LEADERSHIP & EXTRACURRICULARS

NeuroScout

2026 – Present

Chief Operating Officer (COO)

Leading operational strategy and cross-functional team coordination at a technology startup, overseeing product development milestones and day-to-day operations alongside full-time academic commitments.